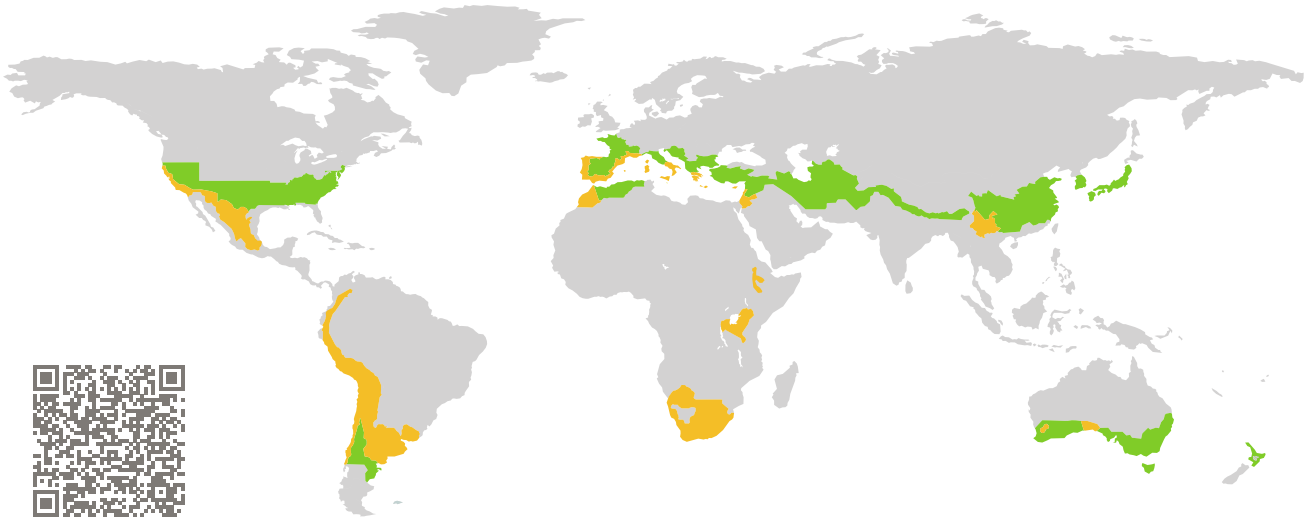
CERTIFICATE

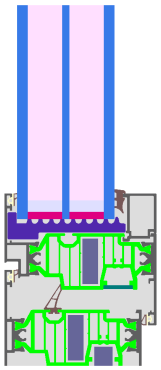
Certified Passive House Component

Component-ID 2570ws04 valid until 31st December 2027

Passive House Institute
Dr. Wolfgang Feist
64283 Darmstadt
Germany



Category:	**Window system**
Manufacturer:	**EVEREST, Tirana, Albania**
Product name:	**EV HYBRID system**



This certificate was awarded based on the following criteria for the warm, temperate climate zone

Comfort	U_W = 1.00	≤	1.00 W/(m² K)
	$U_{W,\text{installed}}$	≤	1.05 W/(m² K)
	with U_g	=	0.90 W/(m² K)
Hygiene	$f_{Rsi=0.25}$	≥	0.65
Airtightness	Q_{100} = 0.15	≤	0.25 m³/(h m)

warm, temperate climate

phB

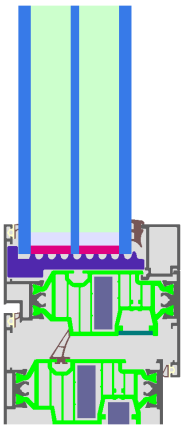
Passive House efficiency class	phE	phD	phC	phB	phA

CERTIFIED COMPONENT

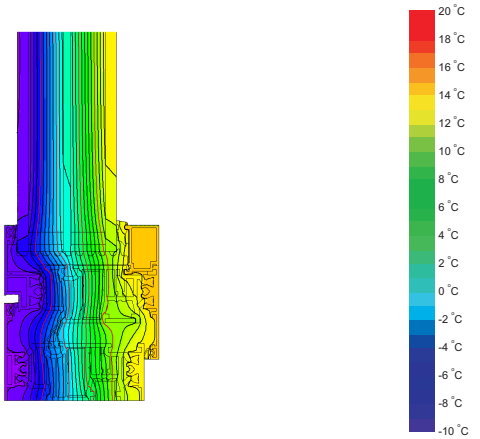
Passive House Institute

www.passivehouse.com

EVEREST
Bulevardi Blu 544, Kamëz, 1030, AL Tirana, Albania
☎ +355 47 20 44 44 | ✉ info@everest.al | http://everest.al |



Calculation model



Isothermal

Description

Aluminium frame with thermal separation made of PVC (0,17 W/(mK)), and insulation insert made of Graphite EPS (0,032 W/(mK)). Maximum dimensions in accordance to the tested parameters. Pane thickness: 54 mm (6/19/4/19/6), rebate depth: 14,25 mm.

Explanation

The window U-values were calculated for the test window size of 2.46 m × 1.48 m with U_g = 0.90 W/(m^2 K). If a higher quality glazing is used, the window U-values will improve as follows:

Glazing	U_g =	0.90	0.70	0.64	0.58	W/(m^2 K)
		↓	↓	↓	↓	
Window	U_W =	1.00	0.85	0.81	0.76	W/(m^2 K)

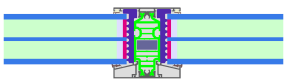
Transparent building components are classified into efficiency classes depending on the heat losses through the opaque part. The frame U-Values, frame widths, thermal bridges at the glazing edge, and the glazing edge lengths are included in these heat losses. A more detailed report of the calculations performed in the context of certification is available from the manufacturer.
The Passive House Institute has defined international component criteria for seven climate zones. In principle, components which have been certified for climate zones with higher requirements may also be used in climates with less stringent requirements. In a particular climate zone it may make sense to use a component of a higher thermal quality which has been certified for a climate zone with more stringent requirements.

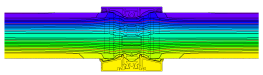
Further information relating to certification
can be found on www.passivehouse.com and passipedia.org.

Frame values		Frame width b_f mm	U-value frame U_f W/(m² K)	Ψ-glazing edge Ψ_g W/(m K)	Temp. Factor $f_{Rsi=0.25}$ [-]
Mullion fixed	(0M1)	72	0.83	0.028	0.75
Mullion 1 casement	(1M1)	115	1.05	0.025	0.74
Mullion 2 casements	(2M1)	157	1.07	0.026	0.74
Door lock side	(DL1)	130	1.07	0.026	0.73
Bottom fixed	(FB1)	53	0.88	0.027	0.75
Top fixed glazing	(FH1)	53	0.88	0.027	0.75
Jamb fixed glazing	(FJ1)	53	0.88	0.027	0.75
Bottom	(OB1)	96	1.04	0.026	0.74
Head	(OH1)	96	1.04	0.026	0.74
Jamb	(OJ1)	96	1.04	0.026	0.74
Threshold	(OT1)	111	1.50	0.025	0.58
Spacer: MULTITECH G			Secondary seal: Silicone		

Mullion
fixed

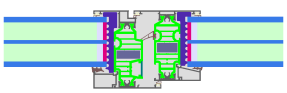
b_f = 72 mm
U_f = 0.83 W/(m² K)
Ψ_g = 0.028 W/(m K)
f_{Rsi} = 0.75

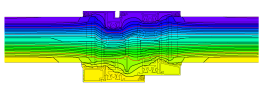




Mullion
1 casement

b_f = 115 mm
U_f = 1.05 W/(m² K)
Ψ_g = 0.025 W/(m K)
f_{Rsi} = 0.74

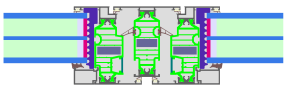
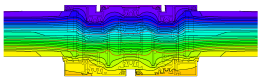




www.passivehouse.com

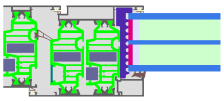
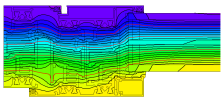
Mullion
2 casements

b_f = 157 mm
U_f = 1.07 W/(m² K)
Ψ_g = 0.026 W/(m K)
f_{Rsi} = 0.74

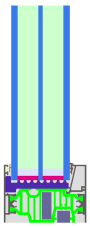
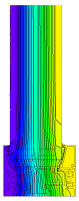
Door
lock side

b_f = 130 mm
U_f = 1.07 W/(m² K)
Ψ_g = 0.026 W/(m K)
f_{Rsi} = 0.73

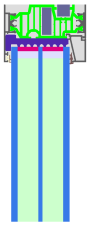
Bottom
fixed

b_f = 53 mm
U_f = 0.88 W/(m² K)
Ψ_g = 0.027 W/(m K)
f_{Rsi} = 0.75

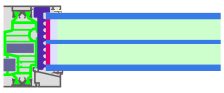
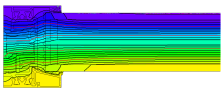
Top
fixed glazing

b_f = 53 mm
U_f = 0.88 W/(m² K)
Ψ_g = 0.027 W/(m K)
f_{Rsi} = 0.75



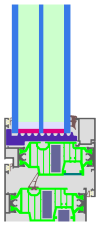

Jamb
fixed glazing

b_f = 53 mm
U_f = 0.88 W/(m² K)
Ψ_g = 0.027 W/(m K)
f_{Rsi} = 0.75

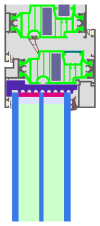
Bottom

b_f = 96 mm
U_f = 1.04 W/(m² K)
Ψ_g = 0.026 W/(m K)
f_{Rsi} = 0.74




Head

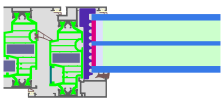
b_f = 96 mm
U_f = 1.04 W/(m^2 K)
Ψ_g = 0.026 W/(m K)
f_{Rsi} = 0.74

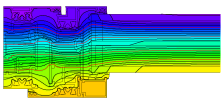




Jamb

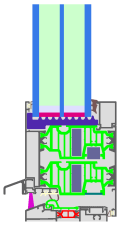
b_f = 96 mm
U_f = 1.04 W/(m^2 K)
Ψ_g = 0.026 W/(m K)
f_{Rsi} = 0.74





Threshold

b_f = 111 mm
U_f = 1.50 W/(m^2 K)
Ψ_g = 0.025 W/(m K)
f_{Rsi} = 0.58

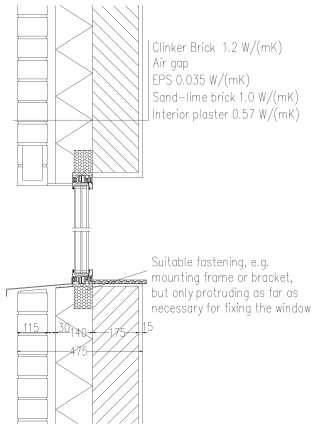




Validated installations

Cavity wall (fixed glazing)

U_{Wall} = 0.22 W/(m² K)

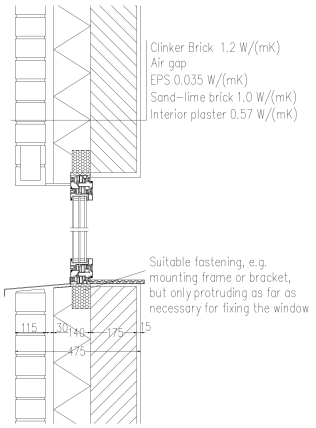


$\Psi_{install}$	W/(m K)
Top	0.017
Left	0.017
Right	0.017
Bottom	0.019

$U_{W,installed}$ = 1.04 W/(m² K)

Cavity wall (operable)

U_{Wall} = 0.22 W/(m² K)

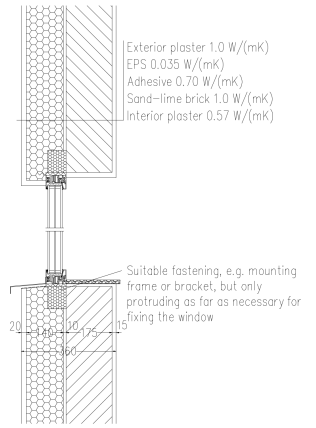


$\Psi_{install}$	W/(m K)
Top	0.016
Left	0.016
Right	0.016
Bottom	0.022

$U_{W,installed}$ = 1.04 W/(m² K)

Exterior insulation and finishing system (EIFS) (fixed glazed)

U_{Wall} = 0.23 W/(m² K)

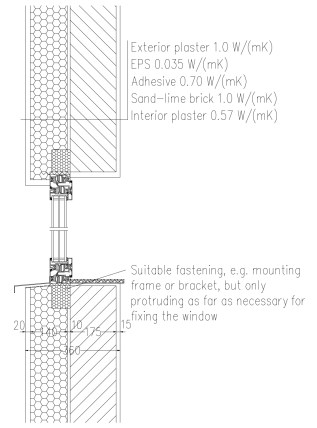


$\Psi_{install}$	W/(m K)
Top	0.013
Left	0.013
Right	0.013
Bottom	0.018

$U_{W,installed}$ = 1.04 W/(m² K)

Exterior insulation and finishing system (EIFS) (operable)

U_{Wall} = 0.23 W/(m² K)

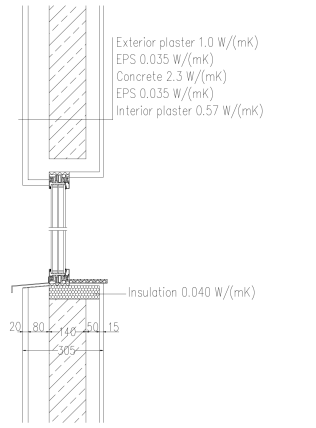


$\Psi_{install}$	W/(m K)
Top	0.015
Left	0.015
Right	0.015
Bottom	0.020

$U_{W,installed}$ = 1.04 W/(m² K)

Formwork blocks (fixed)

U_{Wall} = 0.25 W/(m² K)

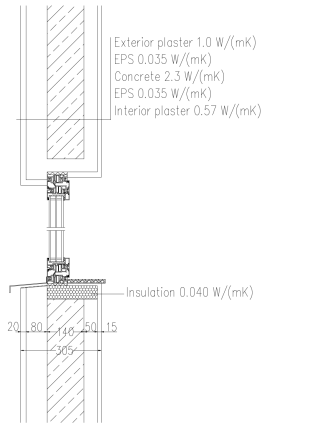


$\Psi_{install}$	W/(m K)
Top	0.014
Left	0.014
Right	0.014
Bottom	0.021

$U_{W,installed}$ = 1.04 W/(m² K)

Formwork blocks (operable)

U_{Wall} = 0.25 W/(m² K)

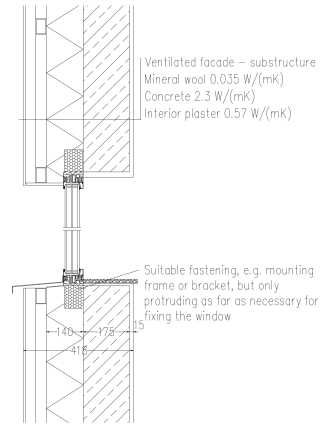


$\Psi_{install}$	W/(m K)
Top	0.014
Left	0.014
Right	0.014
Bottom	0.022

$U_{W,installed}$ = 1.04 W/(m² K)

Ventilated facade (fixed glazed)

U_{Wall} = 0.23 W/(m² K)

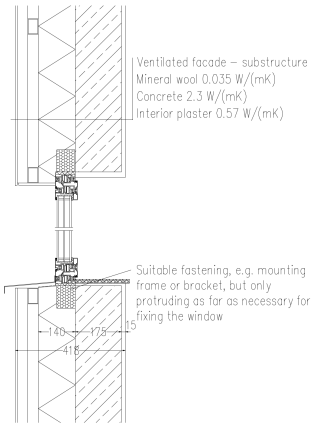


$\Psi_{install}$	W/(m K)
Top	0.014
Left	0.014
Right	0.014
Bottom	0.018

$U_{W,installed}$ = 1.04 W/(m² K)

Ventilated facade (operable)

U_{Wall} = 0.23 W/(m² K)

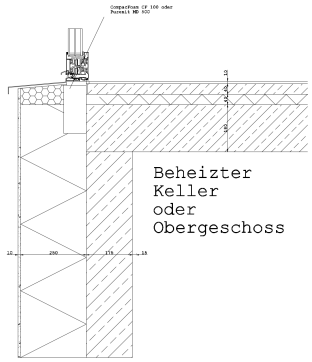


$\Psi_{install}$	W/(m K)
Top	0.016
Left	0.016
Right	0.016
Bottom	0.021

$U_{W,installed}$ = 1.04 W/(m² K)

Exterior insulation and finishing s (EIFS) threshold (operable)

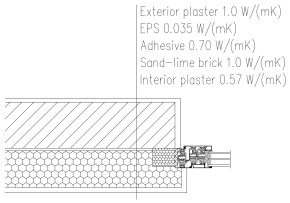
U_1 = 0.23 [W/(m² K)]



$\Psi_{install}$ = 0.052 W/(m K)

Exterior insulation and finishing system (EIFS) jamb (operable)

U_1 = 0.23 [W/(m² K)]



$\Psi_{install}$ = 0.014 W/(m K)

Disclaimer: The Passive House Institute GmbH (PHI) conducts heat-transfer analyses in accordance with the standards set out in Criteria and Algorithms for Certified Passive House Components: Transparent Building Components and Opening Elements in the Building Envelope, based on information provided by the manufacturer. PHI does not verify on-site implementation. It is the responsibility of the project leader to ensure that installed components match the certified specifications in terms of geometry, configuration, and materials. Manufacturers must make full product information available upon request to parties involved in a construction project. These parties may compare the provided information with project documentation and perform on-site inspections as part of the quality-assurance process.